

OPTI-CROP PROJECT DELIVERABLE

Project Planning & Timeline

1. Project Phases & Milestones

The development of Opti-Crop was executed over a 6-week timeline, divided into focused sprint deliverables.

W1: Brainstorming & Requirements
W2: Data Cleaning & ML Training
W3: Database Setup & Backend Development
W4: UI Implementation & Styling
W5: Integration & Functional Testing
W6: Production Deployment & Documentation

2. Task Allocation

- **Ideation & Requirement Analysis (Week 1):** Scoped user needs, drafted DFDs, compiled solution functional specifications.
- **Data Wrangling & Model Engineering (Week 2):** Preprocessed crop recommendation data from Kaggle, handled phosphorus outliers, trained Logistic Regression classifier, and extracted cluster insights.
- **Database & Business Logic Integration (Week 3):** Developed relational tables, integrated models with Flask web controllers, and implemented user session hashing.
- **Front-End Styling & UI Polish (Week 4):** Created a premium glassmorphic visual layout, styled dashboards, and ensured responsive grid layouts.
- **Testing & Quality Assurance (Week 5):** Conducted performance checks, load tests, and verified prediction accuracy metrics.
- **Deployment & Documentation (Week 6):** Relocated files into structured directories, verified Render build commands, and generated documentation guides.